

Clasa a 8-a / Formule si Teoreme Esentiale

I. Ecuatia de Gradul II

Forma: $ax^2+bx+c=0$ ($a \neq 0$)

Discriminant: $\Delta = b^2 - 4ac$

Solutii: $x_{1,2} = (-b \pm \sqrt{\Delta}) / (2a)$

- $\Delta > 0$: doua solutii reale distincte
- $\Delta = 0$: o solutie dubla: $x = -b/(2a)$
- $\Delta < 0$: nicio solutie reala

Relatiile lui Viete: $x_1+x_2 = -b/a$; $x_1 \times x_2 = c/a$

- Ecuatia cu rad. x_1, x_2 : $x^2 - (x_1+x_2)x + x_1 \times x_2 = 0$

II. Sisteme de Ecuatii

- Substitutia: exprimam o necunoscuta, inlocuim in cealalta ecuatie
- Reducerea: inmultim ecuatiile cu factori, adunam/scadem
- 0 solutie: drepte secante; Nicio solutie: paralele; Infinit: coincid

III. Inecuatii de Gradul II

- Rezolvam ecuatia asociata $\Rightarrow$ radacinile $x_1 \leq x_2$
- $a > 0$: $ax^2+bx+c > 0$ pentru $x < x_1$ sau $x > x_2$ (exterior)
- $a > 0$: $ax^2+bx+c < 0$ pentru $x_1 < x < x_2$ (interior)
- $a < 0$: semne inversate fata de $a > 0$
- $\Delta < 0, a > 0$: expresia mereu pozitiva; $\Delta < 0, a < 0$: mereu negativa

IV. Functii

- Functia $f: A \rightarrow B$ asociaza fiecarui x din A exact un y din B

Proportionalitate: $y=kx$ (dreapta prin origine 0)

Gradul I: $y=mx+n$ (dreapta; m =panta, n =ord. la origine)

Gradul II: $y=ax^2+bx+c$ (parabola)

- $m > 0$: functie crescatoare; $m < 0$: descrescatoare
- $a > 0$: parabola U; $a < 0$: parabola cap in jos

Varf parabola: $xV = -b/(2a)$; $yV = f(xV)$

V. Calcul Algebric - Formule remarcabile

$$(a+b)^2 = a^2 + 2ab + b^2; \quad (a-b)^2 = a^2 - 2ab + b^2$$

$$(a+b)(a-b) = a^2 - b^2$$

$$(a+b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$$

$$(a-b)^3 = a^3 - 3a^2b + 3ab^2 - b^3$$

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2)$$

$$a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

VI. Geometrie in spatiu

$$\text{Cub: } A = 6a^2; \quad V = a^3$$

$$\text{Cuboid: } A = 2(ab + bc + ca); \quad V = abc$$

$$\text{Cilindru: } A_{\text{lat}} = 2\pi rh; \quad V = \pi r^2 h$$

$$\text{Con: } A_{\text{lat}} = \pi r l \quad (l = \text{generatoarea}); \quad V = \frac{\pi r^2 h}{3}$$

$$\text{Sfera: } A = 4\pi r^2; \quad V = \frac{4}{3}\pi r^3$$

$$\text{Piramida: } V = \frac{A_{\text{baza}} \times h}{3}$$

$$\text{Trunchi con: } V = \frac{\pi h (R^2 + Rr + r^2)}{3}$$

VII. Trigonometrie

$$\sin A = \frac{\text{cat.opp}}{\text{ipot}}; \quad \cos A = \frac{\text{cat.al}}{\text{ipot}}; \quad \tan A = \frac{\text{cat.opp}}{\text{cat.al}}$$

$$\sin^2 A + \cos^2 A = 1 \quad (\text{identitate fundamentala})$$

$$\tan A = \frac{\sin A}{\cos A}$$

- $30^\circ: \sin = 1/2, \cos = \sqrt{3}/2, \tan = \sqrt{3}/3$
- $45^\circ: \sin = \sqrt{2}/2, \cos = \sqrt{2}/2, \tan = 1$
- $60^\circ: \sin = \sqrt{3}/2, \cos = 1/2, \tan = \sqrt{3}$

$$\text{Teorema sinusurilor: } \frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = 2R$$

$$\text{Teorema cosinusurilor: } a^2 = b^2 + c^2 - 2bc \cos A$$

VIII. Geometrie plana - Recap

Suma unghiuri triunghi = 180° ; patrulater = 360°

Pitagora: $c^2 = a^2 + b^2$; Triplete: (3,4,5)(5,12,13)(8,15,17)

Arie triunghi = $b \times h / 2 = (1/2)ab \sin C$

Cerc: $C = 2\pi r$; $A = \pi r^2$

Lungime arc = $(L/360) \times 2\pi r$; Arie sector = $(L/360) \times \pi r^2$

IX. Vectori si coord. carteziene

Distana AB = $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

Mijlocul [AB]: $M = ((x_1 + x_2)/2, (y_1 + y_2)/2)$

Panta dreptei AB: $m = (y_2 - y_1)/(x_2 - x_1)$

Ecuatia dreptei prin A(x1,y1) cu panta m: $y - y_1 = m(x - x_1)$

- Drepte paralele: $m_1 = m_2$; Perpendiculare: $m_1 \times m_2 = -1$*

X. Probabilitati si Statistica

$P(A) = \text{card}(A) / \text{card}(\Omega)$; $0 \leq P(A) \leq 1$

$P(A \cup B) = P(A) + P(B) - P(A \cap B)$

- Evenimente independente: $P(A \times B) = P(A) \times P(B)$*

Media = $\sum (x_i \times f_i) / N$; Mediana = val. din mijloc

XI. Calcul Combinatoric (notiuni)

Aranjamente: $A(n,k) = n! / (n-k)!$

Combinari: $C(n,k) = n! / (k!(n-k)!)$

Permutari: $P(n) = n!$

- $0! = 1$; $n! = 1 \times 2 \times 3 \times \dots \times n$*

Clasa a 8-a / Exerciții de Recapitulare

Ecuatii de gradul II:

- $x^2 - 5x + 6 = 0$: $a = ______$, $\Delta = ______$, $x_1 = ______$, $x_2 = ______$
- $2x^2 - 8 = 0$: $x = ______$; $x^2 + 2x + 5 = 0$: $\Delta = ______$, solutii = $______$
- Viete pt $x^2 - 7x + 12 = 0$: $x_1 + x_2 = ______$, $x_1 \times x_2 = ______$

Functii:

- $y = 2x - 3$: $m = ______$, $n = ______$, taie Ox in $______$, Oy in $______$
- $y = x^2 - 4x + 3$: varf $V(______, ______)$, taie Oy = $______$, taie Ox = $______$
- Intersectia $y = x + 1$ si $y = x^2 - x - 2$: punctele $______$

Trigonometrie:

- In triunghi dreptu., ipot = 10, $\angle A = 30^\circ$: cat.opp = $______$, cat.al = $______$
- $\sin^2 45 + \cos^2 45 = ______$; $\tan 60 \times \cos 60 = ______$

Geometrie in spatiu:

- Cub $a = 4$: $A = ______$, $V = ______$
- Cilindru $r = 3$, $h = 5$: $V \approx ______$
- Sfera $r = 6$: $A \approx ______$

Inecuatii si sisteme:

- $x^2 - 4 < 0$: solutia $______$; $x^2 - 5x + 6 > 0$: solutia $______$
- $2x + y = 7$ si $x - y = 2$: $x = ______$, $y = ______$